

Topic Modeling of Mpox-related Instagram Posts: Understanding Public Perception over Time

1st Sunipun Seemanta

Department of Computer Science
American International University-Bangladesh (AIUB)
Dhaka 1229, Bangladesh
22-47547-2@student.aiub.edu

2nd Riya Das

Department of Computer Science
American International University-Bangladesh (AIUB)
Dhaka 1229, Bangladesh
22-46592-1@student.aiub.edu

4th M. Mostafizur Rahman

Department of Mathematics
American International University-Bangladesh (AIUB)
Dhaka 1229, Bangladesh
mostafiz.math@aiub.edu

1st Mahmudul Haque Shakir

Department of Computer Science
American International University-Bangladesh (AIUB)
Dhaka 1229, Bangladesh
21-45016-2@student.aiub.edu

3rd Md. Saef Ullah Miah

Department of Computer Science
American International University-Bangladesh (AIUB)
Dhaka 1229, Bangladesh
saef@aiub.edu

5th Mufti Mahmud

Department of Information and Computer Science
SDAIA-KFUPM Joint Research Center for AI
Interdisciplinary Research Center for Bio Systems and Machines
King Fahd University of Petroleum and Minerals
Dhahran, 31261, KSA
mufti.mahmud@kfupm.edu.sa

Abstract—Mpox is now a pandemic disease, and all its public health concerns have inspired very heated debates on Instagram. This paper is trying to capture possible public perceptions, concerns, and narratives through their Instagram posts on Mpox by applying two cutting-edge unsupervised topic models called BERTopic and Latent Dirichlet Allocation (LDA). A dataset of 60127 multilingual posts from July 2022 to October 2024 was processed and raised some clear thematic issues and vaccination hesitance, including disease awareness, awareness of misinformation, and geographic differences. BERTopic investigated nuanced location-specific issues such as humor on social media and technical risks; LDA gave a more extensive framework centered on broader topics like health emergencies and worldwide impact. The base of the analysis is a dataset of Mpox-related Instagram posts collected over a specified time. Pre-processing consists of text extraction and removal of URLs, tokenization, and removal of stop words, and also uses countVectorization. These findings serve to shed light on the interaction between public sentiment and health communication and the critical importance of customized outreach programs to counter misinformation and enhance awareness and public health response in outbreaks of infectious diseases. Findings yield insights into public health communication in setting proactive plans to curb misinformation and boost awareness, with future studies along the lines of cross-platform analysis as well as with multi-media data touching on the greater public perception.

Index Terms—Topic Modeling, Mpox Monkeypox, Machine learning, Deep learning, Instagram Post.

I. INTRODUCTION

Mpox is an infectious disease caused by the monkeypox virus, and it is mostly spread by person-to-person close contact

or, less commonly, through contaminated objects or surfaces. Zoonotic transmission may occur with animals through close contact with infected wildlife or their products [1]. The monkeypox virus was first noted in late 1958 among monkeys. In 1970, it was confirmed that the virus affected humans in the Democratic Republic of the Congo. The virus then became an international health concern as outbreaks around the world took place, including in the U.S. in 2003 and a significant resurgence in 2022, which drove the WHO to declare a Public Health Emergency [2]. Monkeypox diagnosis is currently performed through real-time PCR testing. Specimens are to be collected from skin lesions and must be assayed by public health or commercial laboratories [3]. Prevention from monkeypox is heavily dependent on protocols for infection control, targeted vaccination of high-risk groups, and continued development of vaccines to assure safety and efficacy [4]. As the number of Mpox patients increased, so did online discussions, especially on popular platforms like Twitter, Reddit, Instagram, TikTok, and others. When people post information, personal stories, and comments on the ongoing situation, these interactions provide qualitative and quantitative information relevant to tracking public concern and mapping the landscape of misinformation and other issues about the disease [5], [6]. Social media platforms capture in real-time the responses to unfolding health events that offer a unique window into public sentiment. It focuses on images, the use of hashtags, and comments by its users. It has become crucial in the evolution of public health communication and

sentiment analysis today [7]. Studying the content and themes of Mpox-related posts on Instagram can help track what issues, myths, and aiding structures are available in the population and changes in qualitative and quantitative awareness during the situation. This research analyzes the patterns of topics that arise to understand the changes in public attitudes towards Mpox due to specific events, health policies, and the media. Therefore, this study aims to enhance the existing public health strategies by mapping out the evolution of social concerns addressed to Mpox. This will help develop better communication plans for more effective responses in case of similar outbreaks.

II. LITERATURE REVIEW

Nirmalya Thakur indicates the growing importance of creating datasets from social media during outbreaks of viruses in this research. However, no prior work has created a multilingual Instagram dataset for the study of sentiments, hate speech, and anxiety around the mpox outbreak [16]. Andy Edinger et al. (2023) has highlighted the importance of real-time monitoring of social media during emergencies to combat the rapid spread of misinformation, such as that which occurred during the mpox outbreak [8]. This research by Yunwen Wang et al. (2024) relies on the analysis of BERTopic on tweets created by self-identified sexual-minority men and gender-diverse individuals around topics like health activism, vaccination, and stigmatization while promoting inclusive health communication and stigma prevention in infectious disease outbreak contexts [10]. Yunwen Wang et al. (2024) demonstrated the application of BERTopic to mpox discourse within sexual minority and gender-diverse individuals, which are highlighted within the study, which underscores the necessity for public health approaches that are inclusive against stigma and inequities [9]. This research by V.S. Anoop and S. Sreelakshmi (2023) applied NLP-based topic modeling and sentiment analysis to Reddit comments to understand public perceptions across themes of stigma, fear, and governmental initiatives during the Mpox outbreak and offered insights that could inform health policy and intervention strategies [11]. V. S. Anoop demonstrated the topic modeling using NLP to public sentiment and views on Mpox-themes such as transmission, government actions, homophobia, and fear and stigma as expressed on social media [12]. Christopher J. McKinley et al. (2024) was the analysis on the Twitter discourse during the mpox outbreak of 2022. Tom May et al. (2023) proposed the need to co-design interventions; this research validates their value in ensuring acceptability and, thus, improved effectiveness in public health messaging, health services, and programs for GBMSM, including mpox prevention and mitigation strategies [13]. This displays a transition where uncertainty immediately followed the announcement by the WHO on public health emergencies towards more risk-focused narratives. It touches on how immediacy in public health messages influences social media conversations [14]. Zhou Tong and Haiyi Zhang (2016) described the purpose of this paper to explore the application of Latent Dirichlet Allocation (LDA) in text mining. The two experiments involved

investigating Wikipedia articles and Twitter users' interests. The paper thus presents an integrated framework for topic modeling and demonstrates the potential use of the same for organizing, searching through, and analyzing huge amounts of textual data and its possible applications in social and business research [15].

III. METHODOLOGY

In this part, it is described from where the data is collected, which type of data is collected, and gives a deeply overview about data set in the data collection section. Then comes the preprocess subsection, where mainly talk about how data was cleaned, stop words were removed, and the data set was pre-processed. and at last, the model selection part indicates which model was selected and why. Fig-1 graphically represents the whole process.

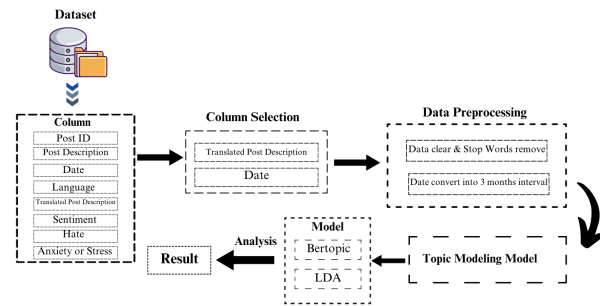


Figure 1: Overview of the research methodology

A. Data Collection

For this research, a Mpox Instagram Dataset was used from an existing paper *Mpox Narrative on Instagram: A Labeled Multilingual Dataset of Instagram Posts on Mpox for Sentiment, Hate Speech, and Anxiety Analysis* [16], Where this dataset has 8 columns such as Post ID, Post Description, Date, Language, Translated Post Description, Sentiment, Hate, Stress or Anxiety. This multilingual (52 languages) data set contains 60,127 Instagram posts. This post was translated using Google API and put into the Translated Post Description column. Figure-2 visually shows date vs. translated post Description From the dataset, where the vertical axis indicates a number of posts and the horizontal axis indicates date. On the other hand The word cloud (figure-3) gives an overview of key terms from translated post descriptions that have something to do with "monkeypox" (mpox). The dominant terms are "monkeypox", "disease", "virus", "spread", "public health", and "symptom", which indicate that the content is majorly talking about health concerns, disease transmission, and public awareness. Other notable words are "emergency", "case", "vaccine", and "Ministry", which shows discussions of health emergencies, confirmed cases, prevention measures, and government or institutional responses. The use of emphatic terms like "people" and "health" reflects the very societal and health effects of the disease. In general, this word cloud

[illegible]

B. Preprocess

2) *Data set preprocess*: For this research, use only two columns from this dataset, translated post description and time. The dataset (Translated Post Description) was divided into three-month intervals. Graphically shown in figure-4.

C. Model Selection

The BERTopic model will be used because previous research [17] has shown that it has better topic separation, clear topic meanings, and helps readers understand the text structure

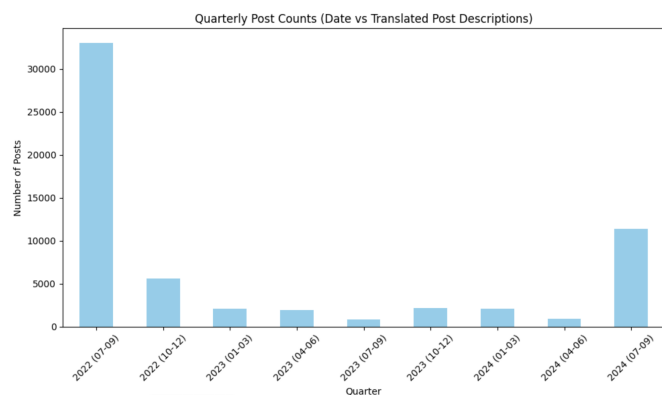


Figure 4: Quarterly Post Count

Latent Dirichlet Allocation (LDA) is one of the most popular in this field [21]. LDA performs significantly well in intra-mixing distributional documents from multiple topics. [22]. This model's algorithm for text is based on statistical (Bayesian) topic models. This helps to sort documents, find information, and suggest content.

IV. RESULT AND DISCUSSION

This section discusses the result and this paper's findings using the Bertopic and LDA models. Figure-5 and table-I show the result of the Bertopic model. The line chart indicates 927 individual topics in Instagram posts about Mpox from July 2022 to October 2024.

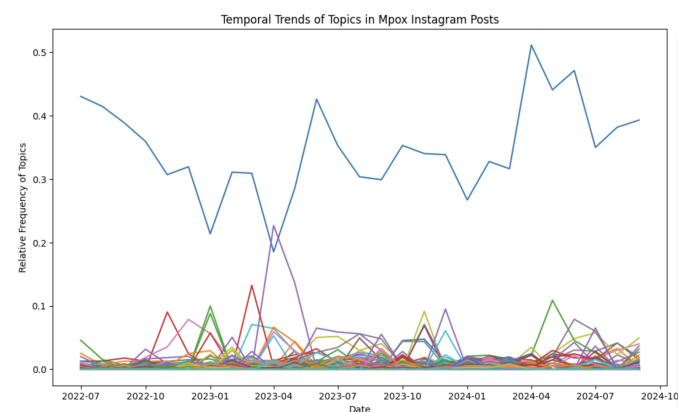


Figure 5: Temporal Trends of Topic in Mpox Instagram Post(Bertopic)

Table I: Summary of Topics with Keywords and Total Posts for BERTopic Analysis

Heading	Topic	Keywords	Total Post
Most Range of Post	Topic -1	virus, monkeypox, symptoms, disease, health, contact, cases, mpox, vaccine, skin	23,034 posts
	Topic 0	global, emergency, health emergency, world health, declared, global health, Congoorganization, health organization, Africa	684 posts
	Topic 1	men, gay, men sex, sex, sex men, bisexual, gay men, gay bisexual, stigma, sexual	671 posts
	Topic 2	monkeypox vaccine, got, vaccine, dose, shot, pox vaccine, dose monkeypox, vaccination done, monkeypox vaccination	577 posts
	Topic 3	Pakistan, Karachi, Islamabad, passengers, Saudi Arabia, airport, Sindh, airports	571 posts
Middle Range of Post	Topic 17	pregnant, women, pregnant women, pregnancy, postpartum, breastfeeding, technical note, birth, fetal	251 posts
	Topic 18	COVID monkeypox, COVID, monkeypox, COVID19, Martine, Marlier, texts, COVIDmemes	248 posts
	Topic 19	fluids, monkeypox spread, respiratory, contact, direct, bodily, clothing, prolonged	247 posts
	Topic 20	surfaces, soap, cleaning, gloves, avoid, wash, soap water, disinfection, hands, washing	242 posts
	Topic 21	monkeypox staysafestayhome, monkeypox, comedy, FYP, LOL, besafe	224 posts
Least Range of Post	Topic 124	emergency, monkeypox following, WHO announcement	61 posts
	Topic 127	freedom, Vaccines, variant, rapid test	59 posts
	Topic 134	AIDS Day, HIV, mental health awareness	40 posts
	Topic 138	word puzzle, Instagram, dictionary	39 posts
	Topic 141	mpox, regional director, Europe	35 posts
Last 3	Topic 916 to 918	Unknown	less than 15 posts

From fig- 5 each line represents a different topic, besides Table-I, which indicates the most, middle range, and least posted topic with a post count and keywords which related mpox. Using Bertopic model this research found that topic 1 focused on general discussion, symptoms, spread, vaccination, and health-related information about monkeypox, and it is the central topic for the community. Topic 0 is about the global health emergency announced and the actions of health organizations. Topic 1 addresses the intersection of monkeypox and its impact on various communities. Topic 2 indicates vaccination campaigns, availability of vaccines, and experiences after receiving the vaccine. Some middle-range topics, like Topic 17 discuss its side effects, risk during pregnancy, etc. However, Mpox spreads and prevention are the central topics of both topics 18 and 19. Topic 20 focuses on preventing mpox through hygiene and cleaning practices. There are topics like 916 and 917 (posts less than 15) that do not indicate a Pacific topic.

On the other hand, figure-6 graphically represents, and table-II shows the tabular representation of the outcome of the LDA model. Find a total 10 types of topics from July 2022 to October 2024 using the LDA model. This model shows the most discussed topic is Topic 8, Topic 7, and Topic 4. Where topic 8 focuses on monkeypox and its global impact. It is the most concerning topic during the outbreak. Topic 7 indicates the symptoms and transmission. Topic 4 highlights specific risk groups and vaccines. Topic 0, Topic 9, and Topic 3 are the medium-discussed topics. Topic 0 indicates people focus on general health tips and are also conscious about monkeypox. Topic-9 links monkeypox with the COVID-19 pandemic and focus on conspiracy theories or doubts. Topic 3 is about cases, but people get less interested over time. Topic 2,5,6 have low engagement for different reasons. Topic 2 focuses on specific people or events, which aren't interesting to a wide audience.

Topic 6 talks about health issues like HIV and cancer, but the connection to monkeypox is weak, so it doesn't attract people looking for monkeypox updates. Topic 5 like war, religion, and monkeypox, which makes it confused and less important for health.

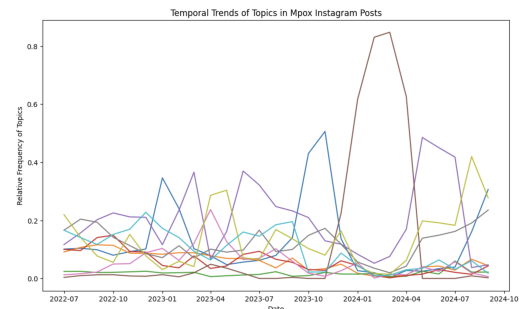


Figure 6: Temporal Trends of Topic in Mpox Instagram Post(LDA)

Table II: LDA Model Output

Heading	Topic No	Keywords	Total Post
Most Discussed Topics	Topic 8	disease, public, outbreak, world, virus, emergency, cases, mpox, health, monkeypox	10,581 posts
	Topic 7	fever, people, health, infected, skin, disease, virus, contact, symptoms, monkeypox	10,408 posts
	Topic 4	sex, risk, men, hiv, people, vaccination, mpox, health, vaccine, monkeypox	8,897 posts
Medium Discussed Topics	Topic 0	healthinfo, don, let, mpox, stay, health, friends, healthy, 39, monkeypox	7,688 posts
	Topic 9	monkey, like, vaccine, everythingisalie, coronavirus, covid_19, covid19, pandemic, covid	6,931 posts
	Topic 3	suspected, according, case, department, disease, state, confirmed, health, cases, monkeypox	4,756 posts
Least Discussed Topics	Topic 2	people, curtis, jr, peffer, robert, states, united, dr, quot, monkeypox	1,311 posts
	Topic 6	naturally, hiv, human, cancer, lgbtq, hpv, herpes, health, herbs, monkeypox	1,761 posts
	Topic 5	elijah, ukraine, war, repentance, holiness, floodedvehicle, russia, monkeypox, youtube, 39	2,538 posts

According to analysis two models, this research find out both model talk about Health and Emergency, Vaccination, Sexual Transmission, and Awareness. Some unique topics, like BertTopic emphasize regional issues (Karachi, Saudi Arabia), Technical and specific groups (pregnant women, fluids, cleaning), and Social media discussions (humor, memes). On the other hand, LDA captures broader health information dissemination, Specific public figures or unrelated items (Curtis, Robert), and External geopolitical elements (Ukraine, war, repentance). Table-III shows the Comparative Analysis of both model.

Table III: Comparative Analysis of BERTopic and LDA Results

Model	Topic/Category	Post Volume
Shared Topics	Disease & Emergency	High
	Vaccination	Medium
	Sexual Transmission	Medium
Unique to BERTopic	Regional Focus	Low
	Pregnancy and Fluids	Medium
	Social Media & Humor	Low
Unique to LDA	Broader Health Info	Medium
	COVID-19 Connections	High
	Geopolitical Elements	Low

This research finds that people are concerned about mpox all over the year. They also focus on symptoms and general disease information. Besides, people are curious about how Mpox spreads across different groups and regions. Their social media discussions also help to understand which population or areas were affected. People also talk about vaccination.

According to this research, people should pay attention to the following exploration areas: misinformation on social media, intersection with climate change, global health equity, and mental health Impacts. Besides, according to P. K. Tiwari [23] awareness camping also create a huge impact for preventing monkey pox.

V. DATA AND CODE AVAILABILITY

This section contains the modeling code.Github Link-
https://shorturl.at/sCQbD

VI. CONCLUSION

Public sentiment and topic trends about the outbreak of Mpox from July 2022 to October 2024 are discussed in this paper using Instagram posts. Through BERTopic and LDA analysis, 927 topics and 10 topics identified the major public concerns with the symptoms, awareness, and regional impacts of the disease. Discussions of Mpox were decidedly heterogeneous during the study period, and conversations about the symptoms and virus transmission were decidedly salient in early 2023. The findings depict the heterogeneity in public discourse: discussions that began on symptoms and modes of virus transmission are now transformed into discussions on vaccination and their impact on global health. The inter-comparative study of BERTopic and LDA is indicative of their complementary strengths in formulating a nuanced and broader understanding of thematic patterns. Insight into this is significant to public health authorities in developing targeted communication strategies with respect to behavior change, misinformation management, and preparedness for emerging outbreaks. This points not only to a general lack of awareness within the public about prevention and medical treatment and vaccination but also to an urgent necessity for pointed public health communication. The insights that this paper derives go a long way in aiding health authorities to effectively communicate more through better outreach strategies that enhance the awareness of the public about such outbreaks in the future.

BERTopic finds out more about the 900 topics, but for this research, sort them into 3 categories: most posts, middle posts, and least posts. Each category contains only 5 topics. So this paper selected only 15 topics per topic, which may have left out small but important issues. This research focuses on only social media platforms.

REFERENCES

- [1] <https://www.who.int/news-room/questions-and-answers/item/mpox>
- [2] Z.-S. Yang et al., "The first case of monkeypox virus infection detected in Taiwan: awareness and preparation," *International Journal of Infectious Diseases*, vol. 122. Elsevier BV, pp. 991–995, Sep. 2022. doi:10.1016/j.ijid.2022.07.051
- [3] <https://www.cdc.gov/mpox/hcp/diagnosis-testing/index.html#:~:text=Mpox%20is%20diagnosed%20using%20real,%20Dspecific%20testing%2C%20is%20needed>
- [4] B. L. Ligon, "Monkeypox: A review of the history and emergence in the Western hemisphere," *Seminars in Pediatric Infectious Diseases*, vol. 15, no. 4. Elsevier BV, pp. 280–287, Oct. 2004. doi:10.1053/j.spid.2004.09.001
- [5] A. Edinger et al., "Misinformation and Public Health Messaging in the Early Stages of the Mpox Outbreak: Mapping the Twitter Narrative With Deep Learning," *Journal of Medical Internet Research*, vol. 25. JMIR Publications Inc., p. e43841, Jun. 06, 2023. doi:10.2196/43841
- [6] Kumkum, Ananna Rashid, Anik Sen, Abdullah Yousuf Noman, Prajukta Majumder, Tze Hui Liew, Nafiz Fahad, Md Saef Ullah Miah, Riadul Islam Rabbi, and Md Jakir Hossen. "From Data to Diagnosis: Applying Machine Learning Model for Reliable Heart Disease Prediction." In *2025 5th Asia Conference on Information Engineering (ACIE)*, pp. 50–56. IEEE, 2025.
- [7] Q. X. Ng, C. E. Yau, Y. L. Lim, L. K. T. Wong, and T. M. Liew, "Public sentiment on the global outbreak of monkeypox: an unsupervised machine learning analysis of 352,182 Twitter posts," *Public Health*, vol. 213. Elsevier BV, pp. 1–4, Dec. 2022. doi:10.1016/j.puhe.2022.09.008
- [8] Edinger A, Valdez D, Walsh-Buhi E, Trueblood JS, Lorenzo-Luaces L, Rutter LA, Bollen J Misinformation and Public Health Messaging in the Early Stages of the Mpox Outbreak: Mapping the Twitter Narrative With Deep Learning *J Med Internet Res* 2023;25:e43841 doi:10.2196/43841
- [9] Wang Y, O'Connor K, Flores I, Berdahl CT, Urbanowicz RJ, Stevens R, Bauermeister JA, Gonzalez-Hernandez G Mpox Discourse on Twitter by Sexual Minority Men and Gender-Diverse Individuals: Infodemiological Study Using BERTopic *JMIR Public Health Surveill* 2024;10:e59193 doi:10.2196/59193
- [10] Wang Y, O'Connor K, Flores I, Berdahl CT, Urbanowicz RJ, Stevens R, Bauermeister JA, Gonzalez-Hernandez G. Health activism, vaccine, and mpox discourse: BERTopic based mixed-method analyses of tweets from sexual minority men and gender diverse (SMMGD) individuals in the U.S. medRxiv [Preprint]. 2024 Mar 19:2024.03.19.24304519. doi:10.1101/2024.03.19.24304519 Update in: *JMIR Public Health Surveill*. 2024 Aug 13;10:e59193. doi:10.2196/59193. PMID: 38562836; PMCID: PMC10984054.
- [11] V.S. Anoop, S. Sreelakshmi, Public discourse and sentiment during Mpox outbreak: an analysis using natural language processing, *Public Health*, Volume 218, 2023, Pages 114–120, ISSN 0033-3506, doi:10.1016/j.puhe.2023.02.018 <https://www.sciencedirect.com/science/article/pii/S0033350623000781>
- [12] Anoop, V. S. (2024). Analyzing Public Concerns on Mpox Using Natural Language Processing and Text Mining Approaches. In A. Natarajan, M. Galety, C. Iwendi, D. Das, & A. Shankar (Eds.), *Intersection of AI and Business Intelligence in Data-Driven Decision-Making* (pp. 309–330). IGI Global Scientific Publishing. <https://doi.org/10.4018/979-8-3693-5288-5.ch011>
- [13] May, T., Towler, L., Smith, L.E. et al. Mpox knowledge, behaviours and barriers to public health measures among gay, bisexual and other men who have sex with men in the UK: a qualitative study to inform public health guidance and messaging. *BMC Public Health* 23, 2265 (2023). <https://doi.org/10.1186/s12889-023-17196-0>
- [14] XiangYu Yan, Zhuo Li, Chunxia Cao, Longxin Huang, Yongjie Li, Xiangbin Meng, Bo Zhang, Maohe Yu, Tian Huang, Jiancheng Chen, Wei Li, Linhui Hao, Dongsheng Huang, Bin Yi, Ming Zhang, Shun Zha, Haijun Yang, Jian Yao, Pengjiang Qian, Chun Kai Leung, Haojun Fan, Pei Jiang & Tiejun Shui. (2024) Characteristics, Influence, Prevention,

and Control Measures of the Mpox Infodemic: Scoping Review of Infodemiology Studies. *Journal of Medical Internet Research* 26, pages e54874. Crossref

- [15] Tong, Zhou & Zhang, Haiyi. (2016). A Text Mining Research Based on LDA Topic Modelling. *Computer Science & Information Technology*. 6. 201-210. 10.5121/csit.2016.60616.
- [16] Thakur, N. (2024, September 9). Mpox Narrative on Instagram: A Labeled Multilingual Dataset of Instagram Posts on Mpox for Sentiment, Hate Speech, and Anxiety Analysis. *arXiv.org*. <https://arxiv.org/abs/2409.05292>
- [17] Gan, L., Yang, T., Huang, Y., Yang, B., Luo, Y. Y., Richard, L. W. C., & Guo, D. (2024). Experimental Comparison of Three Topic Modeling Methods with LDA, Top2Vec and BERTopic. In *Communications in Computer and Information Science* (pp. 376–391). https://doi.org/10.1007/978-981-99-9109-9_37
- [18] Grootendorst, M. (2020). BERTopic: Leveraging BERT and c-TF-IDF to Create Easily Interpretable Topics. *Zenodo*.doi:10.5281/zenodo.4430182
- [19] Egger, R., & Yu, J. (2022). A Topic Modeling Comparison Between LDA, NMF, Top2Vec, and BERTopic to Demystify Twitter Posts. *Frontiers in Sociology*, 7. <https://doi.org/10.3389/fsoc.2022.886498>
- [20] Sánchez-Franco, M. J., and Rey-Moreno, M. (2022). Do travelers' reviews depend on the destination? An analysis in coastal and urban peer-to-peer lodgings. *Psychol. Market.* 39, 441–459. doi:10.1002/mar.21608
- [21] Jelodar, H., Wang, Y., Yuan, C., Feng, X., Jiang, X., Li, Y., & Zhao, L. (2018). Latent Dirichlet allocation (LDA) and topic modeling: models, applications, a survey. **Multimedia Tools and Applications*, 78*(11), 15169–15211. <https://doi.org/10.1007/s11042-018-6894-4>
- [22] Farkhod, A., Abdusalomov, A., Makhmudov, F., & Cho, Y. I. (2021). LDA-Based Topic Modeling Sentiment Analysis using Topic/Document/Sentence (TDS) model. **Applied Sciences*, 11*(23), 11091. <https://doi.org/10.3390/app112311091>
- [23] P. K. Tiwari, R. K. Rai, S. Khajanchi, R. K. Gupta, and A. K. Misra, "Dynamics of coronavirus pandemic: effects of community awareness and global information campaigns," *The European Physical Journal Plus*, vol. 136, no. 10, Oct. 2021, doi: 10.1140/epjp/s13360-021-01997-6.